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Bispectrum-based technique to remove cross-terms in quadratic systems and Wigner–Ville distribution

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Abstract
In the analysis of multicomponent signals using quadratic nonlinear transforms or systems, along with the desired auto-terms unwanted cross-terms are also generated. In this paper, a novel bispectrum-based technique is proposed to remove these cross-terms. The concept of accumulated modified bispectrum (AMB) is developed. The cross-term removal property of the AMB is demonstrated, and its use in quadratic system analysis in presence of additive noise is illustrated. The application of AMB in interference reduction in the analysis of multicomponent non-stationary signals using Wigner–Ville distribution (WVD) is illustrated. For performance comparison, we compute an information theoretic measure, the Reńyi entropy, which indicates that the performance of the proposed method is better than the smoothed pseudo-WVD in the analysis of multicomponent signals that have same support in time or frequency domain.

Keywords Quadratic systems · Time-frequency analysis · Accumulated modified bispectrum · Wigner–Ville distribution

1 Introduction

Quadratic nonlinear systems and quadratic transforms like the Wigner–Ville distribution (WVD) generate cross-terms in the multicomponent signal analysis. The Wigner–Ville time-frequency distribution is often preferred for time-frequency analysis of non-stationary signals compared to the linear methods like the short-time Fourier transform or Wavelet transform due to the optimal time-frequency resolution property of the WVD [1]. However, the generation of cross-terms in the analysis of multicomponent signals makes it difficult to interpret the results properly, and in the presence of noise, this task becomes even more difficult [1].

There are different methods to remove the cross-terms, and many of these methods are based on designing of some 2-D kernel in the ambiguity domain [2–4]. In these methods, due to time- and frequency-smoothing (filtering), the resolution of the TFRs will not be the same as the resolution of the WVD [5]. One of the solution for dealing with the smoothing effect is the time-frequency reassignment method [6]. The other methods to reduce cross-terms based on signal decomposition are presented in [7,8]. Typically, these methods require non-overlapping time or frequency support for each component.

There are methods that use reference time-frequency distribution as an image for de-noising or cross-term removal in the analysis of multicomponent non-stationary signals [9–12]. Typically, they use spectrogram, pseudo-WVD (PWVD) or smoothed pseudo-WVD (SPWVD) as a reference time-frequency image (TFI) and use different image processing algorithms for de-noising or cross-terms removal purpose. The image processing technique has been applied on the TFI obtained by the wavelet transform of a multicomponent signal to separate components and to perform component-wise signal analysis in [9,10].

In this paper, we propose a new bispectrum-based method to remove the cross-terms resulted due to the quadratic nonlinear nature of a system or transform [13,14]. In the proposed method, a modified form of the bispectrum definition is used. The modified bispectrum gives the peaks at the auto-components. This is achieved by using accumulation of the bispectrum. We illustrate the applications of the developed method in removing cross-terms in the quadratic systems with additive noise and in the analysis of non-stationary signals using WVD. In the analysis of non-stationary signals, each component of the signal is assumed to have the same support either in time or frequency.

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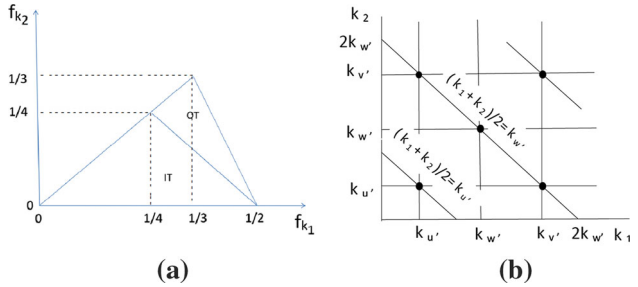


Fig. 1 Modified bispectrum computation. **a** Bispectrum principal region and **b** triple products in modified bispectrum

Since this method is based on the higher-order statistics, it has many advantages like elimination of zero-mean Gaussian noise. The results of cross-term removal by the proposed method are compared with that of the SPWVD in the sequel.

Organization of the paper is as follows: in Sect. 2, we discuss bispectrum definition and its computational details. In Sect. 3, the concept of accumulated modified bispectrum is formulated. In Sect. 4, the applications of the developed concept are illustrated and conclusion is included in Sect. 5.

2 Bispectrum definition and computation

Let us assume that the discrete-time signal $x[n]$, $n = 0, 1, \ldots, N-1$ has the discrete Fourier transform $X[k]$, $k = 0, 1, \ldots, N-1$, where the discrete frequency f_k is expressed as $f_k = \frac{k}{N}$.

The bispectrum of the discrete-time signal $x[n]$ can be computed as

$$B[k_1, k_2] = \mathcal{E}\left\{X[k_1]X[k_2]X^{\star}[k_1 + k_2]\right\} \tag{1}$$

where $\mathcal{E}\{\cdot\}$ is the expectation operator and * denotes complex conjugation. It is not necessary to compute the bispectrum for all values of k_1 and k_2, since it has many symmetries [15,16]. The principal domain of the discrete bispectrum of a band limited signal is shown in Fig. 1a, where $f_{k_1} = \frac{k_1}{N}$ and $f_{k_2} = \frac{k_2}{N}$. The principal region is further divided into two triangular regions called as the inner triangle (IT) and the outer triangle (OT) which are marked in the figure. We assume that the signal is sufficiently oversampled so that there is no aliasing while computing the bispectrum. Hence, it is sufficient to calculate the bispectrum over the inner triangle IT which is defined by $0 \le f_{k_2} \le f_{k_1}$, $f_{k_1} + f_{k_2} \le 1/2$. For the definition of the bispectrum in (1), the computational region of Fig. 1a is appropriate for the real as well as the complex analytic signals [17].

A high value of the bispectrum indicates that the frequency coupling may be present among the source components. Further examination of the phase terms reveals whether the signal components are quadratically phase coupled.

3 Accumulated modified bispectrum

In this section, we modify the definition of the bispectrum, and using the concept of accumulated bispectrum, a method to detect and remove the cross-terms generated due to quadratic nonlinear nature of the systems is presented.

The bispectrum computed using (1) detects quadratic phase coupling. It will be nonzero if $X(k_1)$, $X(k_2)$ and $X(k_1 + k_2)$ are simultaneously nonzero. This occurs when $f_{k_3} = f_{k_1} + f_{k_2}$ and $\phi_3 = \phi_1 + \phi_2$ and these relations imply that the wave at f_{k_3} is a result of quadratic phase coupling between f_{k_1} and f_{k_2}. Here, in this work, we are not interested in the quadratic phase coupling. We want to detect and remove the cross-terms which occur between signal components due to the quadratic nature of the system. To detect the cross-terms, let us consider the modified form of the bispectrum definition

$$B_1[k_1, k_2] = \mathcal{E}\left\{X[k_1]X[k_2]X^{\star}[k_3]\right\} \tag{2}$$

where $k_3 = (k_1 + k_2)/2$.

Note that for the slice $k_2 = k_1$, this becomes

$$\begin{aligned} B_1[k_1, k_2] &= \mathcal{E}\left\{X[k_1]X[k_1]X^{\star}[k_1]\right\} \\ &= \mathcal{E}\left\{|X[k_1]|^3 \angle X[k_1]\right\} \end{aligned} \tag{3}$$

Thus, the condition $k_2 = k_1$, which is a diagonal slice of the modified bispectrum (2), gives the information about the cube of the signal magnitude with linear signal phase. The computed bispectrum contains both the desired auto-terms and non-desired cross-terms. Therefore, the bispectrum slice resulting due to the condition $k_2 = k_1$ is not useful in removing the cross-terms. Hence, we process the signal by computing the modified bispectrum for the case $k_1 \neq k_2$.

Let us assume that $k_{w'} = (k_{u'} + k_{v'})/2$ and $\phi_{w'} = (\phi_{u'} + \phi_{v'})/2$. As shown in Fig. 1b, the modified bispectrum will be nonzero at the locations shown by the dark circles. If we remove the triple products located on the diagonal slice, we will get peaks at the locations $(k_{u'}, k_{v'})$ and $(k_{v'}, k_{u'})$ due to components $k_{u'}, k_{v'}$ and $k_{w'}$.

Thus, the modified bispectrum computed for the case $k_1 \neq k_2$ gives the peaks at the bi-frequencies (f_{k_1}, f_{k_2}) and (f_{k_2}, f_{k_1}) if $f_{k_3} = (f_{k_1} + f_{k_2})/2$ and $\phi_3 = (\phi_1 + \phi_2)/2$. Here, the modified bispectrum phase will be

$$\phi_{B1}[k_1, k_2] = \phi_{k_1} + \phi_{k_2} - \phi_{\left(\frac{k_1+k_2}{2}\right)}. \tag{4}$$

To make the phase of bispectrum B_1 zero in the case of $f_{k_3} = (f_{k_1} + f_{k_2})/2$ and $\phi_3 = (\phi_1 + \phi_2)/2$, we can modify the phase to

$$\phi_{B1}[k_1, k_2] = \phi_{k_1} + \phi_{k_2} - \phi_{\left(\frac{k_1+k_2}{2}\right)} - \phi_c \tag{5}$$

where $\phi_c = \phi_{(k_1+k_2)/2}$ [18].

Note that in the computation of the modified bispectrum, we do not compute it for the condition $k_1 = k_2$, but practically to avoid leakage effects, we need to select some ϵ_t around diagonal slice such that abs $(k_1 - k_2) < \epsilon_t$, and compute the modified bispectrum for the frequency points excluding this set.

Further, we compute the modified bispectrum for the integer values of discrete frequency variable $k_3 = (k_1 + k_2)/2$, which indicates that we can compute it for either both k_1, k_2 even or both odd. In the proposed applications, since we are using the accumulated bispectrum, this computation gives satisfactory results. Nevertheless, if required, we can use either curve fitting or interpolation to construct the modified bispectrum values at the points where $(k_1 + k_2)/2$ is not an integer.

We propose the accumulation of the modified bispectrum to convert the information present in the modified bispectrum to 1-D. We compute the accumulated modified bispectrum (AMB) using

$$\mathrm{AMB}[k_1] = \sum_{\substack{k_2 = 0 \\ k_2 \neq k_1}}^{(N-1)-k_1} B_1[k_1, k_2] \tag{6}$$

where $B_1[k_1, k_2]$ is the modified bispectrum computed in the region defined by $f_{k_1}, f_{k_2} \geq 0$ and $f_{k_1} + f_{k_2} \leq \frac{1}{2}$. In this way, we convert the 2-D information to 1-D. The accumulation of the modified bispectrum will give peaks at f_{k_1} and f_{k_2} if $f_{k_3} = (f_{k_1} + f_{k_2})/2$ and $\phi_3 = (\phi_1 + \phi_2)/2$. Thus, the AMB gives the auto-terms present in the signal and the cross-terms are removed in the AMB.

4 Interference reduction in WVD using AMB

In this section, we develop an application of the AMB in reducing the cross-terms in the WVD of multicomponent signals. Each component of these signals is assumed to have same support either in time or frequency.

It should be pointed out that in this work, we are computing the bispectrum of a squared signal (or TFI) to detect any cross-term present in the signal (or image) and to remove it in the process. The use of bispectrum for the above application is very different compared to its application for extracting various features of any signal (or image). Note that we are computing the bispectrum of the WV transformed data here, which are not the original given data.

Let the base signal $x_0(t)$ be a linear chirp, $x_0(t) = e^{j(2\pi F_0 t + \frac{1}{2}\beta t^2)}$ and let $x_1(t) = x_0(t - t_1)e^{j2\pi F_1 t}$ and $x_2(t) = x_0(t - t_2)e^{j2\pi F_2 t}$ be the time and frequency shifted versions of the base signal. Let us form a composite signal $x(t) = x_1(t) + x_2(t)$, then, the WVD of $x(t)$ is given by

$$\begin{aligned} WV_x(t, F) = {} & \delta(F - (F_1 + F_0) - \beta(t - t_1)) \\ & + \delta(F - (F_2 + F_0) - \beta(t - t_2)) \\ & + 2\delta(F - (F_m + F_0) - \beta(t - t_m)) \\ & \times \cos(2\pi F_d(t - t_m) - t_d(F - F_m) + F_d t_m) \end{aligned} \tag{7}$$

where $\delta(F)$ is the Dirac delta function, $F_d = (F_2 - F_1)$, $t_d = (t_2 - t_1)$, $F_m = \left(\frac{F_1+F_2}{2}\right)$, and $t_m = \left(\frac{t_1+t_2}{2}\right)$.

Let us assume that the multicomponent signal has the same time support, i.e., $t_d = t_1 - t_2 = 0$. The above WVD can be written as

$$\begin{aligned} WV_x(t, F) = {} & \delta(F - (F_1 + F_0) - \beta(t - t_1)) \\ & + \delta(F - (F_2 + F_0) - \beta(t - t_1)) \\ & + 2\delta(F - (F_m + F_0) - \beta(t - t_1)) \\ & \times \cos(2\pi F_d(t - t_1) + F_d t_1) \end{aligned} \tag{8}$$

This indicates that the closely spaced signal components will have small F_d, so the oscillation rate of the cross-term amplitude will be small. In this work, we consider the WVD as a 2-D signal like image and call it as a time-frequency image (TFI). We denote the TFI by $I[n, k]$.

Let us take a slice of the above TFI by fixing discrete-time variable n at instantaneous time n_i,

$$\begin{aligned} I[k; n_i] = {} & \delta[f_k - (f_1 + f_0) - \beta n_i] \\ & + \delta[f_k - (f_2 + f_0) - \beta n_i] \\ & + A_1\delta[f_k - (f_m + f_0) - \beta n_i]|_{f_k = k/N}, \\ & k = 0, 1, \ldots, N - 1 \end{aligned} \tag{9}$$

where A_1 is the amplitude of the cross-term. The first two terms are the auto-components, and the third term in (9) is a cross-term.

Now, we compute the AMB of the signal in (9). The AMB of (9) will be

$$\begin{aligned} \mathrm{AMB}[k; n_i] = {} & A_1\,\delta[f_k - (f_1 + f_0) - \beta n_i] \\ & + A_1\,\delta[f_k - (f_2 + f_0) - \beta n_i] \end{aligned} \tag{10}$$

Thus, the AMB retains the auto-components of the signal and cross-components are eliminated from the composite signal. This computation is repeated for each n_i.

Similarly, for the signals that have the same support in frequency domain, it can be shown that the cross-components can be eliminated using AMB-based method.

Thus, the proposed method comprises of the following steps:

1. Compute the WVD of the multicomponent signal. This time-frequency distribution is taken as an image called the TFI for further processing.
2. In the resulting TFI, calculate the AMB of each row or column as stated below.
3. In selection of dimension row or column for computing the AMB, the time and frequency support of the signal is considered. We select the dimension row/column that has the same support for all the components and compute the AMB for each row or column of the TFI.
4. Once the AMB is computed, to get the exact magnitude of the distribution containing only auto-terms, we derive the mask which will be 1, if the AMB is greater than the threshold and zero at other locations. Let us call this mask as $Y'[k; n_i]$. Here, the chosen threshold is the mean value of the AMB$[k; n]$.
5. Compute the auto-components of the signal as

$$Y_1[k; n_i] = \mathrm{WV}[k; n_i] \,.\, Y'[k; n_i] \tag{11}$$

where $\mathrm{WV}[k; n_i]$ is the Wigner–Vile distribution of the signal at $n = n_i$ and an element-to-element multiplication is used on the right-hand side of the above equation.

In all examples given below, we consider the component signals being aligned in one of the dimensions of the TFI, and we take this dimension to compute the AMB. If the component signals are not aligned along any axis of the TFI, then we need to find an axis of alignment, and compute the AMB along this axis. The procedure can be carried out by rotating the coordinate system.

5 Simulation results

In this section, we present the simulation results of applications of AMB in removing cross-terms in the magnitude spectrum of different stationary and non-stationary signals. In Example 1, we consider the squaring system and illustrate how the AMB can be used in removing cross-terms resulting from the quadratic nature of the system.

In Examples 2, 3 and 4, we will illustrate the AMB-based method for cross-terms removal in the WVD of the multicomponent non-stationary signals. The details of the applications are presented in the following subsections.

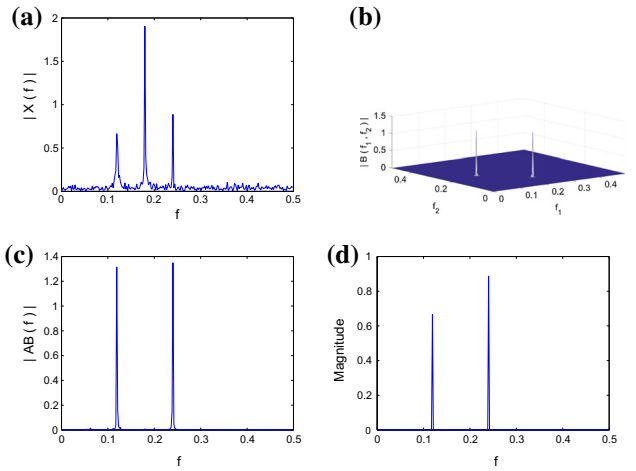


Fig. 2 Cross-terms removal of the signal in Example 1. **a** Magnitude spectrum, **b** modified bispectrum, **c** accumulated modified bispectrum and **d** auto-terms of the signal

5.1 Cross-terms removal in quadratic nonlinear systems

We consider the sum of complex sinusoidal signals passing through the squaring system. We detect and remove the undesired cross-terms at the output of the quadratic system.

Example 1 Consider the sum of complex sinusoidal signals passed through the squaring system as

$$x[n] = \left(\sum_{i=1}^{M} e^{j(2\pi f_i n)} \right)^2 + w[n] \tag{12}$$

where $M = 2$, $f_1 = 0.06$, $f_2 = 0.12$, and $w[n]$ is zero-mean white Gaussian noise at the system output. We set SNR = 6 dB, number of data points $N = 512$, and the number of realizations used in computation of the bispectrum is 500. The value of ϵ_t used is 10. The resulting output signal will contain the auto-terms at $f_1 = 0.12$ and $f_2 = 0.24$, and a cross-term at $f = 0.18$. In Fig. 2a, the output signal magnitude spectrum shows these components.

The modified bispectrum computed for the output signal is shown in Fig. 2b. Clearly, two peaks can be observed which are at bi-frequencies (0.12, 0.24) and (0.24, 0.12). The accumulated bispectrum as given by (6) is computed and plotted in Fig. 2c. The plot is showing strong peaks only at the auto-components, and the cross-components are absent. We create the mask using the procedure described in Sect. 4 and obtain the exact magnitudes of the auto-terms of the signal spectrum as

$$Y_1(f) = |X(f)| \cdot Y'(f) \tag{13}$$

where $|X(f)|$ is the magnitude spectrum of the original signal, $Y'(f)$ is the derived mask, and an element-to-element multiplication is used on the right-hand side of the above equation. As shown in Fig. 2d, the cross-component is removed from the magnitude spectrum.

Note that in case of real sinusoids, we will get one additional term at $f_2 - f_1$. For removing the cross-terms in the AMB, an additional constraint of $f_2 \neq 2f_1$ where $f_2 > f_1$ will be required. Nevertheless, this component can be detected and discarded using the concept of quadratic nonlinearity power index (QNLPI) spectrum [19].

5.2 Removal of cross-terms in WVD using AMB

In the following examples, we apply the AMB-based method as developed above to the WVD of non-stationary signals. In these applications, we use the deterministic version of the bispectrum defined in (2). For illustration purpose, we consider non-stationary signals with two components. However, our method works for multicomponent signals with more than two components. The first signal comprises of the sum of complex chirp components with different chirp rates. In this signal, each chirp component has the same time support, but has different frequency support. In the next example, we consider the signal with the same frequency support for its constituent components but with different time supports, and in Example 4, we discuss the case of the signal with constituent components having the same time support and with overlapping frequency support. Finally, we analyze the vowel sound /ee/ from the spoken word 'deed' using WVD and AMB.

Example 2 Consider the sum of complex linear chirp signals as

$$y[n] = \sum_{i=1}^{M} e^{j(2\pi f_{oi} n + \pi f_{ri} n^2)}, \quad n = 1, \ldots, N \tag{14}$$

where $M = 2$, $f_{o1} = (190/800)$, $f_{r1} = (-30/800^2)$, $f_{o2} = (75/800)$, $f_{r2} = (50/800^2)$, and $N = 512$. In computation of the AMB, the value of ϵ_t used is 40.

The result of applying method of AMB on the signal in Example 2 can be seen in the following figures. The WVD is shown in Fig. 3a. This is taken as TFI for further processing. The AMB computed for each frequency domain signal obtained for every column of the TFI is shown in Fig. 3b. The WVD after removing the cross-term using the procedure stated in Sect. 4 is shown in Fig. 3c.

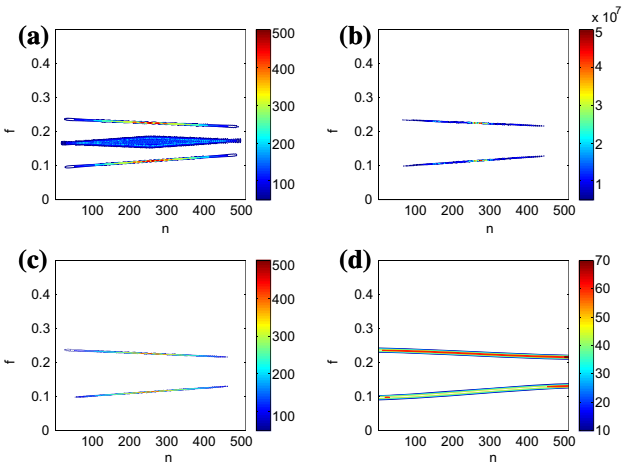


Fig. 3 Cross-terms removal in Example 2 using AMB. **a** WVD of the signal, **b** AMB of the signal, **c** auto-terms of the signal and **d** SPWVD of the signal

The smoothed pseudo-WVD (SPWVD) is defined by

$$\mathrm{SPWV}_y(t, F) = \int_{-\infty}^{\infty} h(t-u) \int_{-\infty}^{\infty} g_1(\tau) y\left(u + \frac{\tau}{2}\right) \times y^{\star}\left(u - \frac{\tau}{2}\right) e^{-j2\pi F\tau} d\tau \, du \tag{15}$$

where $g_1(\tau) = g(\tau/2)g^{\star}(-\tau/2)$, $g(\tau)$ and $h(\tau)$ are frequency-smoothing and time-smoothing windows, respectively [1]. This TFR smooths the WVD in both the time and frequency directions, so it causes smearing in both the time and frequency directions but removes most of the cross-terms.

The SPWVD computed using time- and frequency-smoothing windows is shown in Fig. 3d. Here, both the time- and frequency-smoothing windows used are Hanning windows with length 90. These window lengths are chosen by trial such that we get the sharpest possible time-frequency distribution without interference. This procedure is followed for all the remaining examples also.

For comparison, we have computed the Reńyi entropy which is defined as

$$\mathrm{H}_{\alpha}(\mathrm{TFR}) = \frac{1}{1-\alpha} \log_2 \times \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{\mathrm{TFR}(t, F)}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathrm{TFR}(u, v) \mathrm{d}u \, \mathrm{d}v} \right)^{\alpha} \mathrm{d}t \, \mathrm{d}F \tag{16}$$

where $\alpha = 3$ is the order of the Reńyi entropy H [20]. The Reńyi entropy measures the information extraction

Table 1 Reńyi entropy H_α(TFR), $\alpha = 3$ for different TFRs

TFR	Example 2	Example 3	Example 4	Example 5
WVD	10.9985	10.5490	10.9916	10.9916
SPWVD	13.4293	11.7358	14.8204	12.2522
AMB-WVD	11.6080	10.8727	11.1000	11.5635

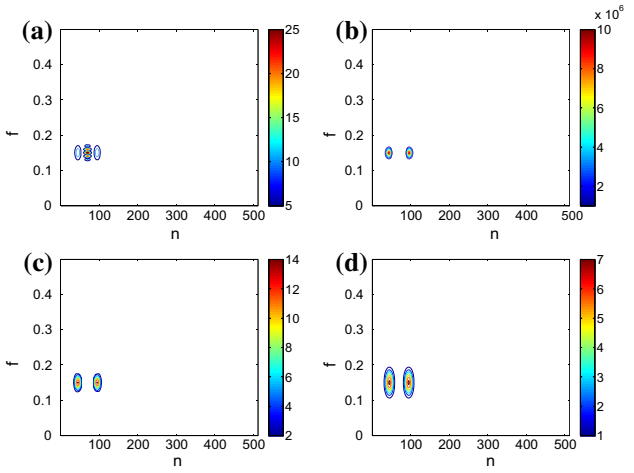


Fig. 4 Cross-terms Removal in Example 3 using AMB. **a** WVD of the signal, **b** AMB of filtered WVD of the signal, **c** auto-terms of the signal and **d** SPWVD of the signal

performance of TFRs. Minimizing the information in a particular TFR is equivalent to maximizing its concentration/resolution.

The computed values of the Reńyi entropy for the WVD, SPWVD and AMB-based method are 10.9985, 13.4293 and 11.6080, respectively, vide Table 1. As can be seen from the Reńyi entropy values and Fig. 3c, d, the sharpness/resolution of the AMB-based method is better than that of the SPWVD.

Example 3 Consider the following signal with limited time and frequency supports

$$y[n] = \sum_{i=1}^{M} A_i \left\{ e^{-\left(\frac{n-n_{0i}}{S_i}\right)^2 \pi} \right\} \left\{ \frac{e^{J(2\pi f_i (n-n_{0i}))}}{e^{J(2\pi f_i n_{0i})}} \right\}, \quad n = 1, \ldots, N \tag{17}$$

where $M = 2$, $A_1 = A_2 = 1$, $f_1 = f_2 = 0.15$, $S_1 = S_2 = 20, n_{01} = 45, n_{02} = 95$ and $N = 512$. In computation of the AMB, the value of ϵ_t used is 40. In the following examples, the TFI is preprocessed with the filter [1 1 1 1 1 1 1 1 1 1] to smooth out the cross-term which oscillates at a frequency proportional to the difference in frequency- and time-shifts of the signals [8]. This step ensures that the proposed method works properly. As shown in Fig. 4a, the WVD of the signal indicates that each component of the signal has the same frequency support, but has different time support.

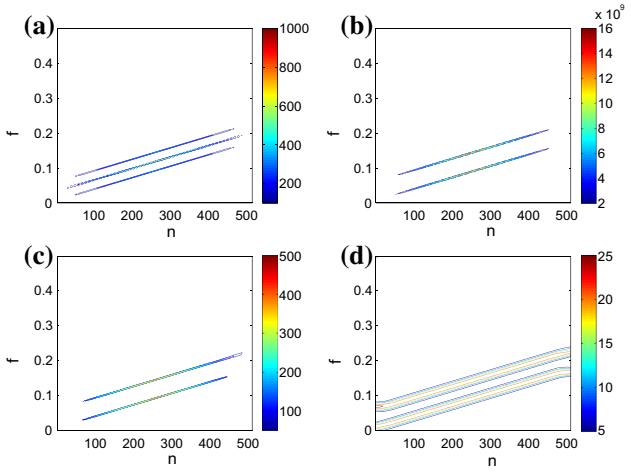


Fig. 5 Cross-terms removal in Example 4 using AMB. **a** WVD of the signal, **b** AMB of filtered WVD of the signal, **c** auto-terms of the signal and **d** SPWVD of the signal

The proposed AMB is computed for each signal obtained for every row of the low-pass filtered TFI. In computation of the AMB, the actual signal values (rows) of the resulting TFI are used in Eq. (2). The computed AMB of the TFI is shown in Fig. 4b. The result shows that the proposed AMB eliminates the cross-term present in the signal. The actual amplitudes of the auto-terms extracted from the WVD are shown in Fig. 4c. The SPWVD of the signal is also shown in Fig. 4d. The time- and frequency-smoothing windows used are Hanning windows with lengths of 21 each. The window lengths are chosen by trial.

The computed values of the Reńyi entropy for the WVD, SPWVD and AMB-based method are 10.5490, 11.7358 and 10.8727, respectively, vide Table 1. Thus the Reńyi entropy values and Fig. 4c, d indicate that the proposed method performs better than the SPWVD.

Note that the amplitude of the cross-term of WVD is oscillatory. To reduce the effect of amplitude oscillation, we used the low-pass filter. Due to this filtering, the TFI becomes smooth, and in turn the AMB and extracted auto-terms also become smooth functions.

Example 4 Consider the sum of parallel complex linear chirp signals as

$$y[n] = \sum_{i=1}^{M} e^{j(\omega_{oi} n + \frac{\gamma_i}{2} n^2)} \tag{18}$$

where $M = 2$, starting angular frequencies $\omega_i = 2\pi f_{oi}$, $f_{o1} = 6/800$, $f_{o2} = 48/800$ and chirp rates $\gamma_i = 2\pi f_{ri}$, $f_{r1} = f_{r2} = 210/800^2$, number of data points $N = 512$ and $\epsilon_t = 40$. The TFI is prefiltered as in Example 3.

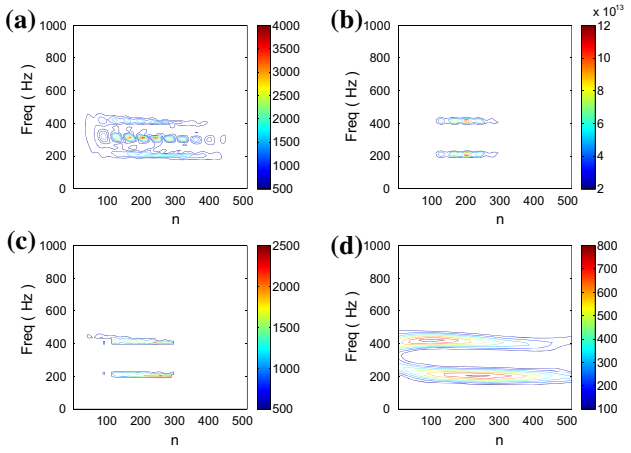


Fig. 6 Analysis of speech vowel /ee/ from word 'deed.' **a** WVD of vowel sound /ee/, **b** AMB of the signal, **c** auto-components and **d** SPWVD of the signal

The WVD of the signal is shown in Fig. 5a where along with two chirp signals the cross-term between two chirps can be seen. It is clear from the WVD that this signal is not separable in time or frequency. Each component of the signal has the same time support and overlapping frequency support. The AMB for each signal obtained from every column of the filtered TFI is computed and is shown in Fig. 5b. We can observe here that the cross-term present in the WVD is removed in the AMB.

The actual amplitude of the auto-terms extracted from the WVD is shown in Fig. 5c. For comparison, the smoothed pseudo-Wigner–Ville distribution (SPWVD) is also computed and is shown in Fig. 5d. The time- and frequency-smoothing windows used are Hanning windows with lengths 71 and 51, respectively. The window lengths are chosen by trial for best performance.

The computed values of the Reńyi entropy for WVD, SPWVD and AMB-based method are 10.9916, 14.8204 and 11.1000, respectively, vide Table 1. Thus, the Reńyi entropy values and Fig. 5c, d indicate that the proposed method resolution is close to the resolution/sharpness of the WVD as compared to the SPWVD.

Example 5 In this example, we consider the natural signal. Here, we analyze the vowel sound /ee/ from the spoken word 'deed.' We are interested in the low-frequency contents of the signal, so we pre-filter the signal with a low-pass filter with cutoff frequency 1500 Hz. In computation of the AMB, the value of ϵ_t used is 9.

The WVD plot of the signal is shown in Fig. 6a. Two significant components along with the cross-term can be seen in the figure. The computed AMB of the WVD is shown in Fig. 6b. The cross-term is removed in the AMB. To extract the original amplitudes of the auto-terms we derive a binary mask by selecting a threshold which is the mean value of the AMB. We have selected this threshold which gives satisfactory performance. The extracted auto-terms can be clearly seen in Fig. 6c. The SPWVD is shown in Fig. 6d. The broadening of the auto-terms is visible in the SPWVD. The smoothing windows chosen are Hanning windows with lengths 201. The window lengths are chosen by trial. The values of the Reńyi entropy for WVD, SPWVD, and AMB-based method are 10.9916, 12.2522 and 11.5635, respectively, vide Table 1.

The values of the Reńyi entropy for all the above examples are listed in Table 1. As can be observed in all examples, the values of Reńyi entropy for the AMB-based method are close to the Reńyi entropy values of the WVD. The lower values of Reńyi entropy indicates the better resolution performance of the method. Thus, the performance of the proposed method is better than the SPWVD method in removing the cross-terms with resolution close to the resolution of the WVD.

6 Conclusion

In this paper, a new bispectrum-based method is proposed to remove the cross-terms generated in the analysis of multicomponent signals employing a quadratic nonlinear system or a quadratic TFR like the WVD. A modified form of the bispectrum definition is proposed for the above purpose. The concept of accumulated bispectrum or modified bispectrum is developed, and computational details are explained. The application of the proposed method in the analysis of multicomponent signals passing through the quadratic nonlinear system is illustrated.

Some applications of the developed concept based on the AMB for cross-term removal in the WVD of a multicomponent non-stationary signal are illustrated. In these applications, we considered various synthetic signals with different supports in time and frequency domains. We also presented an analysis of the vowel sound /ee/ using the AMB and WVD.

The computed measure Reńyi entropy and simulation results indicate that the proposed method performs better than the SPWVD in the analysis of various multicomponent signals. It shows that the concentration/resolution of the AMB-based method is better than that of the SPWVD. In fact, we get nearly optimal resolution of the WVD when the proposed AMB-based method is applied for removal of the cross-terms.

It should be emphasized that unlike the SPWVD method, the AMB-WVD method is based on utilizing the inherent properties of the signal. Thus, the proposed method provides near optimal performance without trial, whereas the SPWVD method always involves trial on window function for reasonable performance.

The Wigner higher-order spectra (WHOS), particularly bispectrum and trispectrum, are computed for extracting time-varying spectral features of a signal [21], and for multicomponent signals, the presence of cross-terms is a crucial problem. Some work related to modifications of defining equations and inclusion of properly designed window function in time or frequency domain to reduce (or completely remove) the cross-term effect has been reported in [22,23]. It will be worth investigating how our method for removal of cross-terms in the WV spectrum (WVD) can be extended to the WHOS without the use of smoothing window.

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